



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 2 Introduction to general pathology Part 2

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Learning Objectives

- ☐ **Know what are general pathologies key stages are .**
- ☐ **Describe what is cellular adaptation and the four main types of it**
- ☐ **Explain cell injury and differentiate between reversible and irreversible cell injury.**
- ☐ **Know the different types of cell necrosis and how they appear and why.**
- ☐ **Understand what apoptosis is.**

Introduction

General Pathology

The process of disease progression can be broken down into several key stages, including **cellular adaptation, injury, death, inflammation, and repair.**

1-Cellular Adaptation, Cells constantly adjust to changes in their environment. Cellular adaptation refers to a cell's response to stress.

- ❑ **Hypertrophy:** An increase in cell size, leading to an increase in organ size.
- ❑ **Hyperplasia:** An increase in the number of cells in an organ or tissue. This can be a physiological response, such as breast glandular tissue growth during pregnancy.
- ❑ **Atrophy:** A decrease in cell size and number, resulting in a reduction in organ size. Muscle atrophy can occur due to disuse.
- ❑ **Metaplasia:** A change where one differentiated cell type is replaced by another. A classic example is the change from ciliated columnar epithelium to stratified squamous epithelium in the bronchi of smokers.



Hypertrophy



Hyperplasia



Atrophy

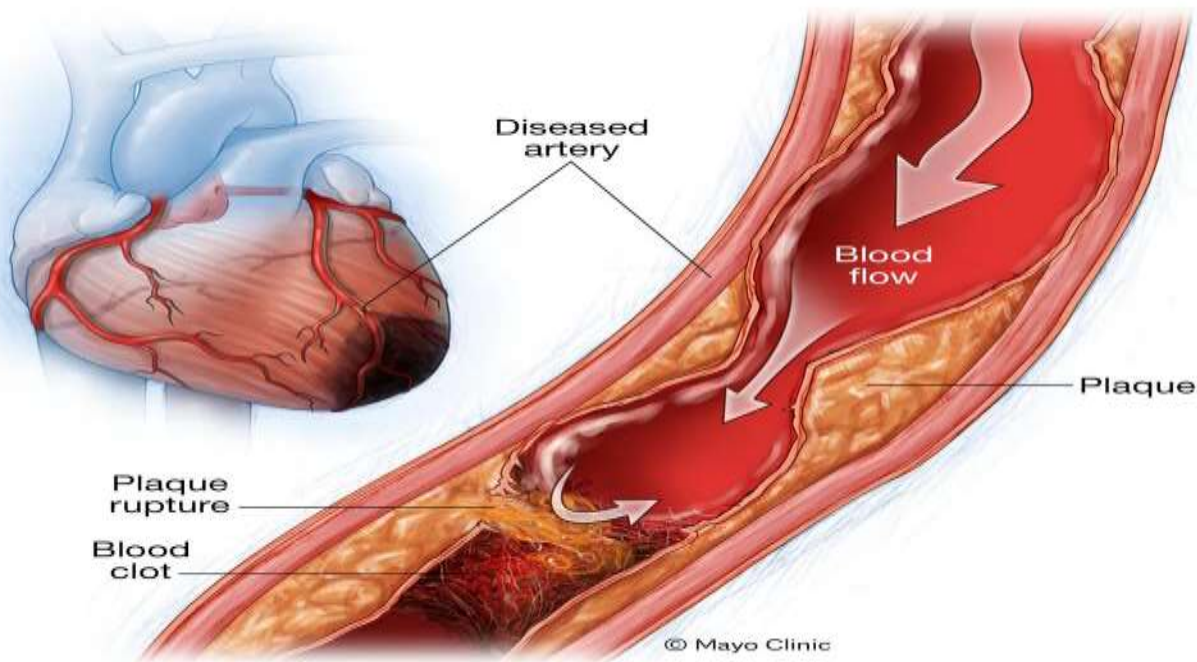


Metaplasia

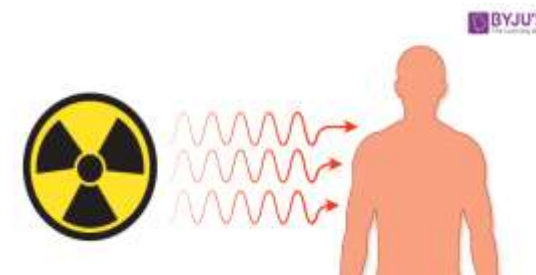
Introduction

2-Cellular Injury, When a cell's adaptive capacity is exceeded or the stress is severe, cellular injury occurs. This can be **reversible** or **irreversible**. The causes of injury (etiology) are broad and include:

A-Hypoxia/Ischemia: Lack of oxygen due to reduced blood flow.



B-Physical agents: Trauma, extreme temperatures, radiation.

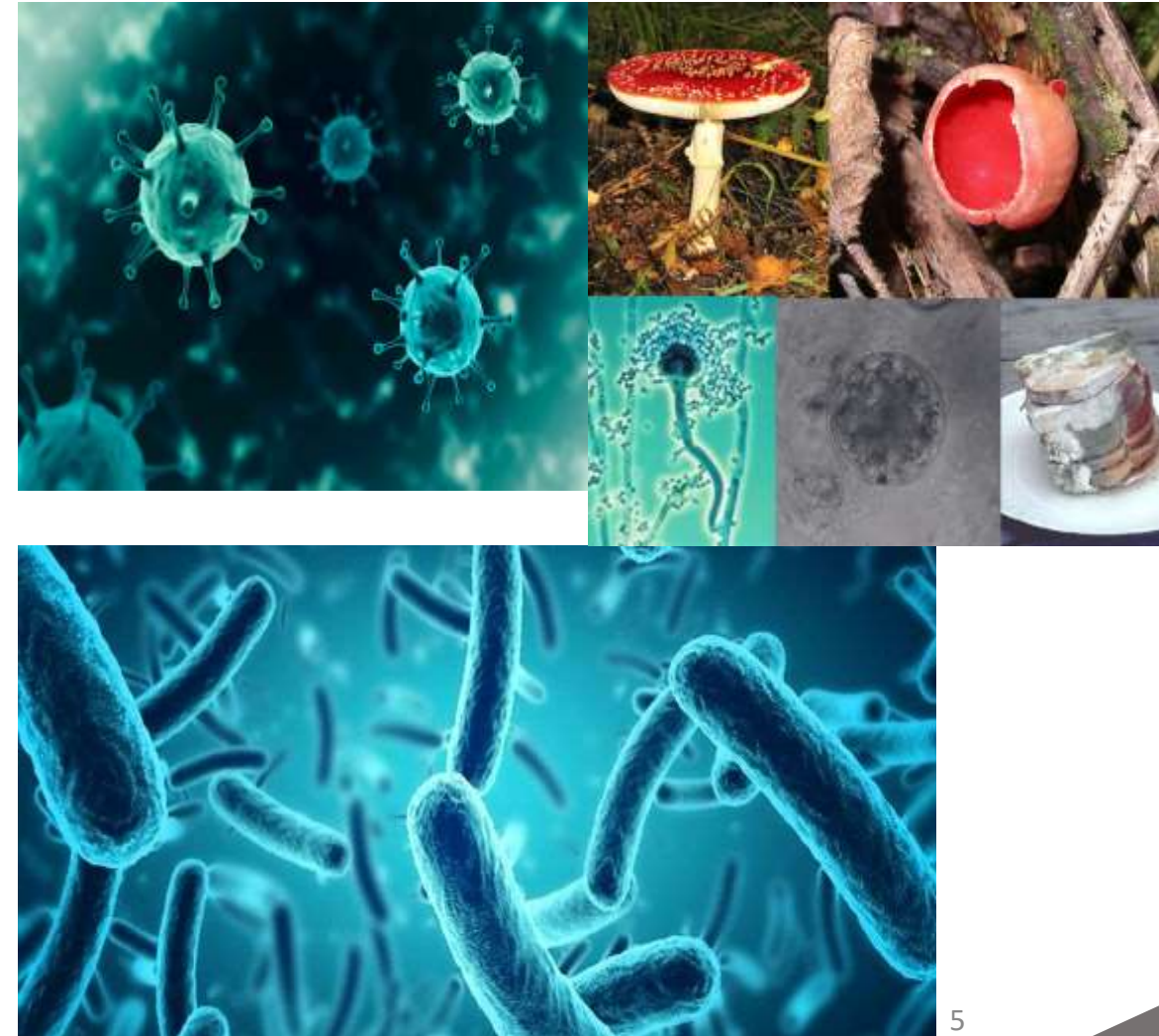


Cellular Injury

C-Chemical agents/Drugs: Poisons, alcohol, medications.



D-Infections: Viruses, bacteria, fungi.

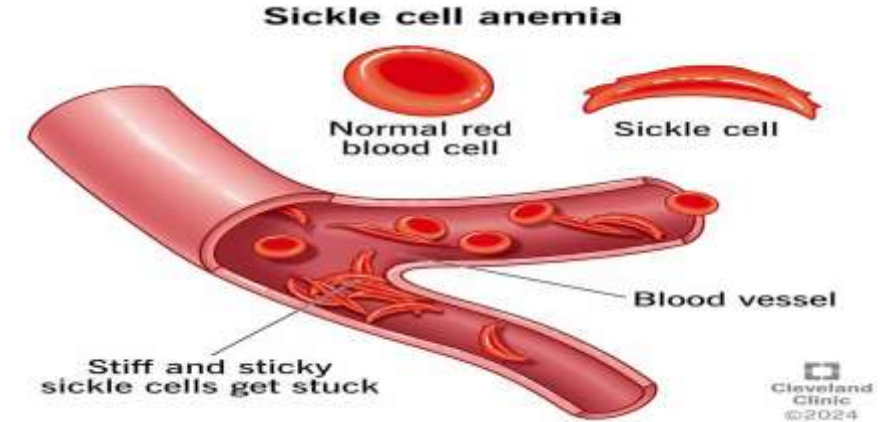


Cellular Injury

E-Immunological reactions: Autoimmune diseases, allergies.

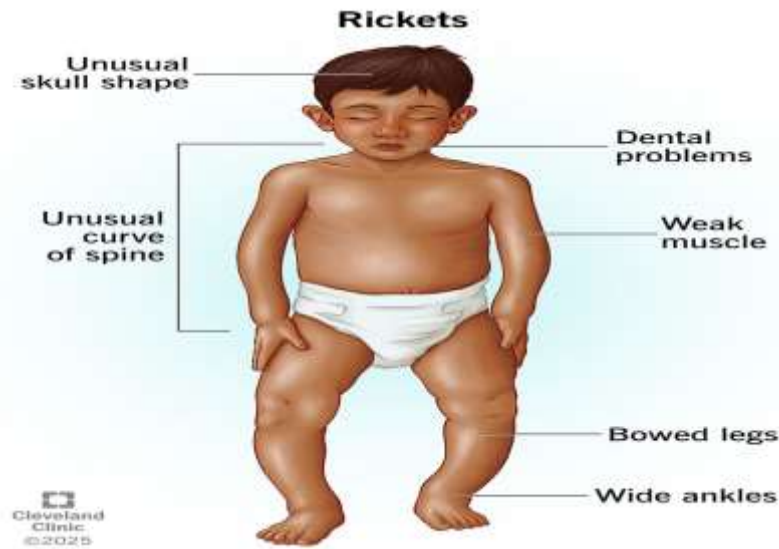


F-Genetic derangements: Inherited diseases.



G-Nutritional imbalances:
Vitamin deficiencies or
excesses.

Rickets(Vitamin D)



Cellular Injury

Types of Cellular injury:

- 1-Reversible
- 2-Irreversible

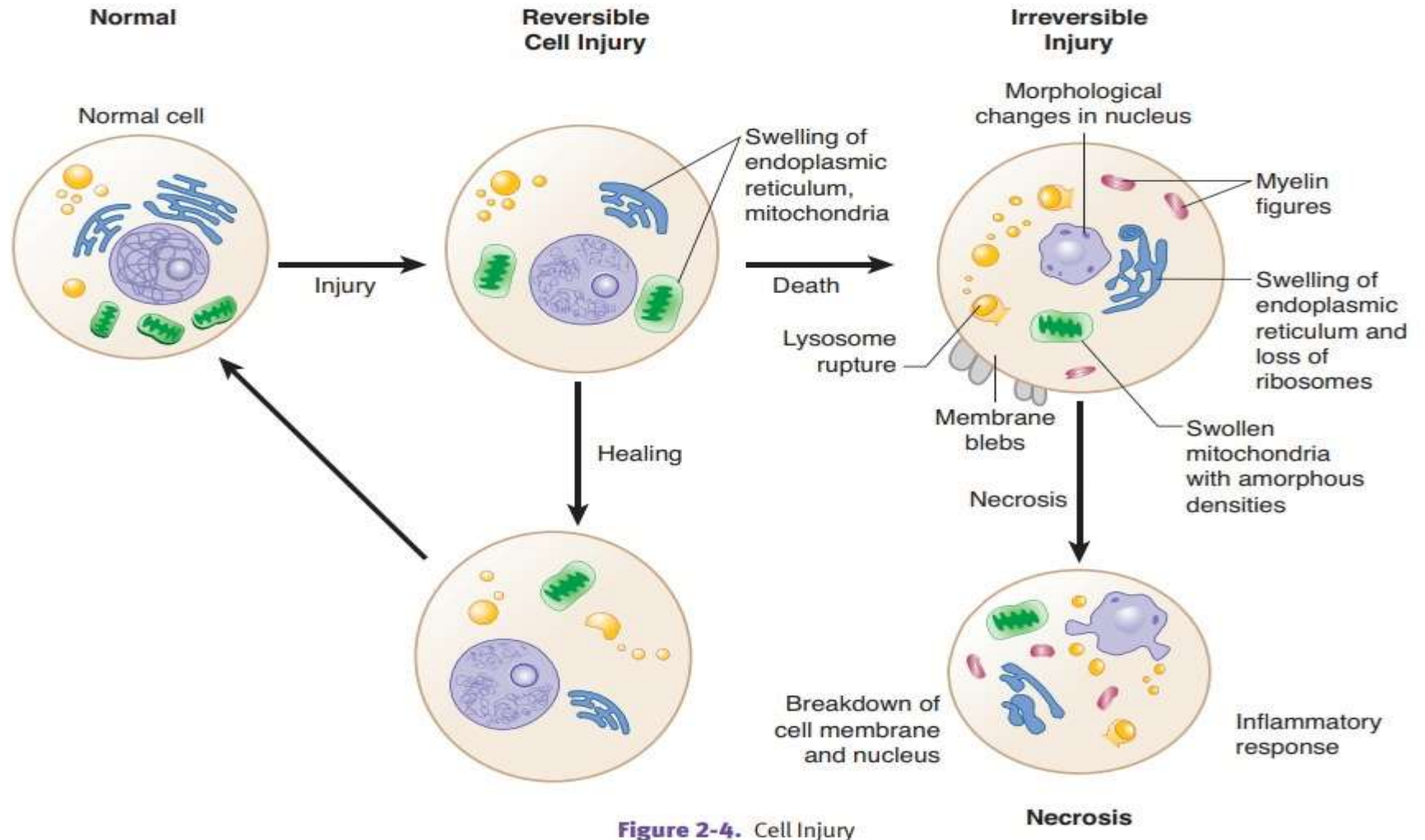
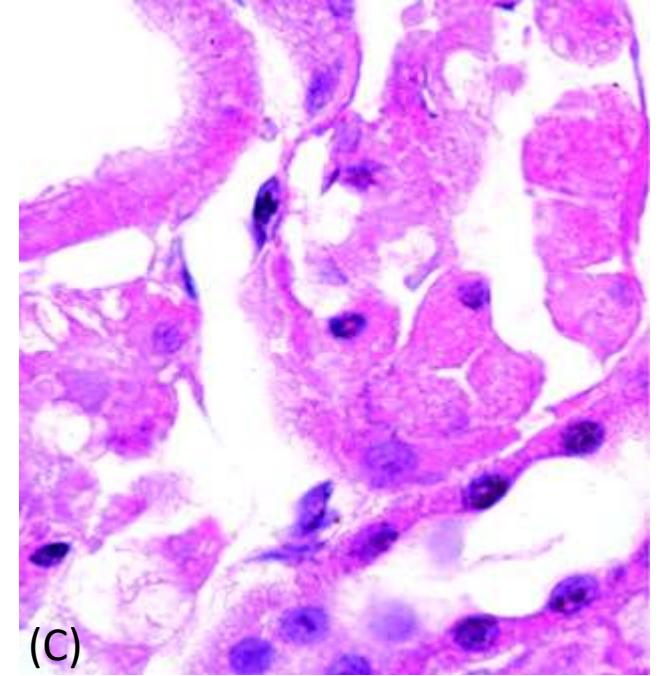
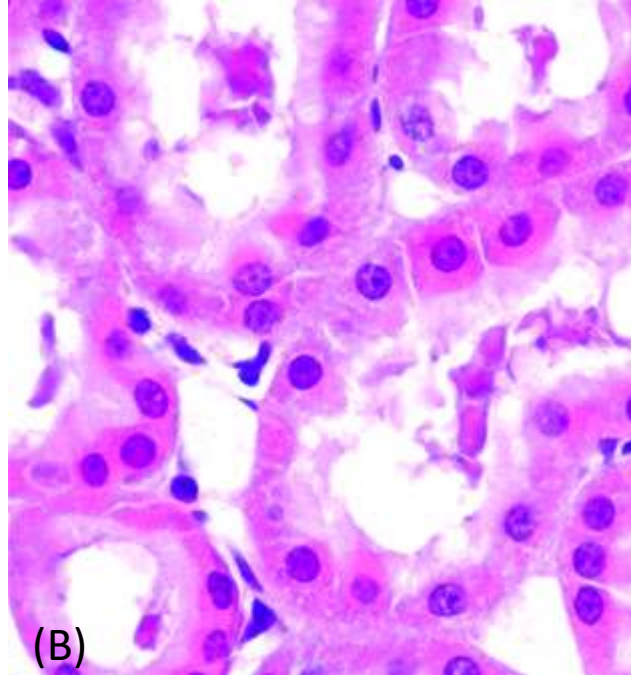
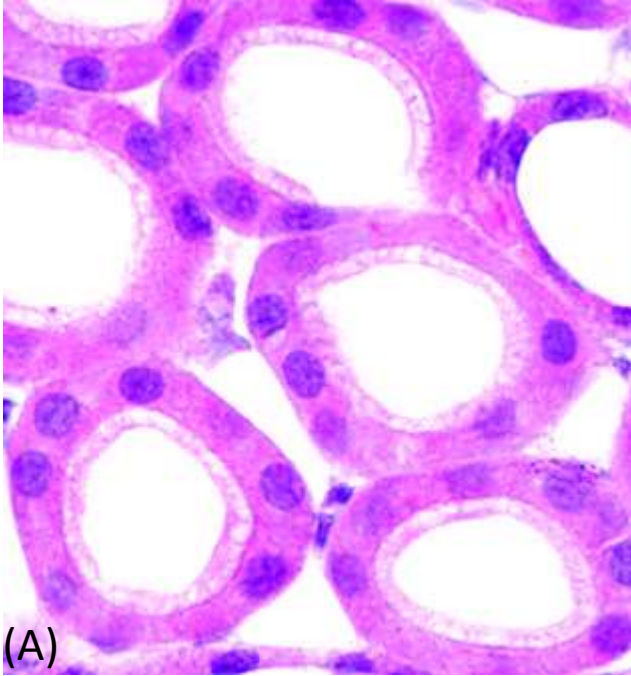


Figure 2-4. Cell Injury

Cellular Injury



(A) Normal kidney tubules: cells look healthy and alive.

(B) Reversible injury (early ischemia): cells start to swell, the surface shows small blebs (bulges), the cytoplasm looks pinker (eosinophilic), but the cells are still alive.

(C) Irreversible injury (necrosis): cells lose their nuclei, break apart, and their contents leak out showing the cells are dead.

Cellular Death

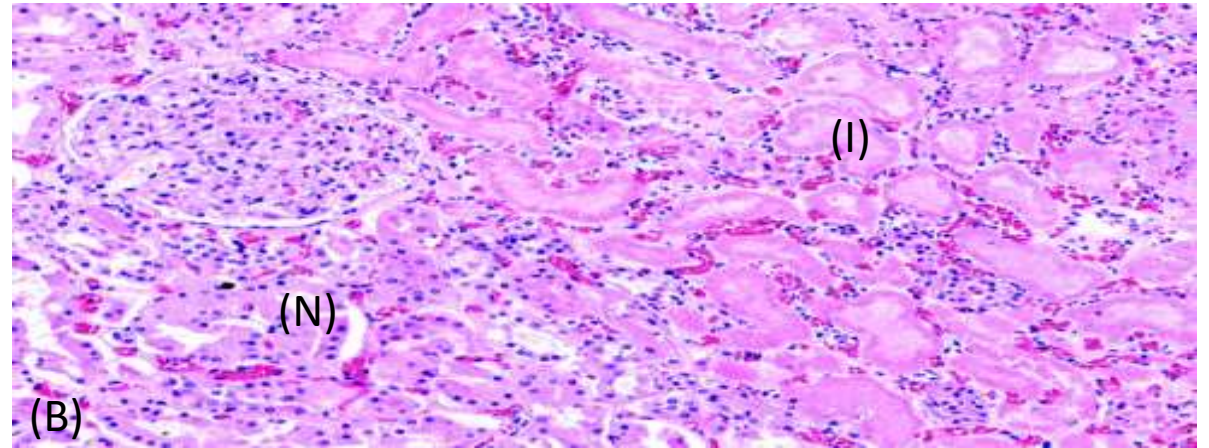
3-Cell Death ,If a cell can't recover from injury, it undergoes cell death. There are two primary forms:

A- Necrosis: A pathological form of cell death caused by severe injury. It involves cell swelling, loss of membrane integrity, and leakage of cellular contents, which triggers a local inflammatory response. Different types of necrosis are named for their gross and microscopic appearance:

- ❑ **Coagulative necrosis:** Occurs in solid organs (e.g., heart, kidney) and is caused by ischemia. The architecture of the tissue is preserved for a few days.



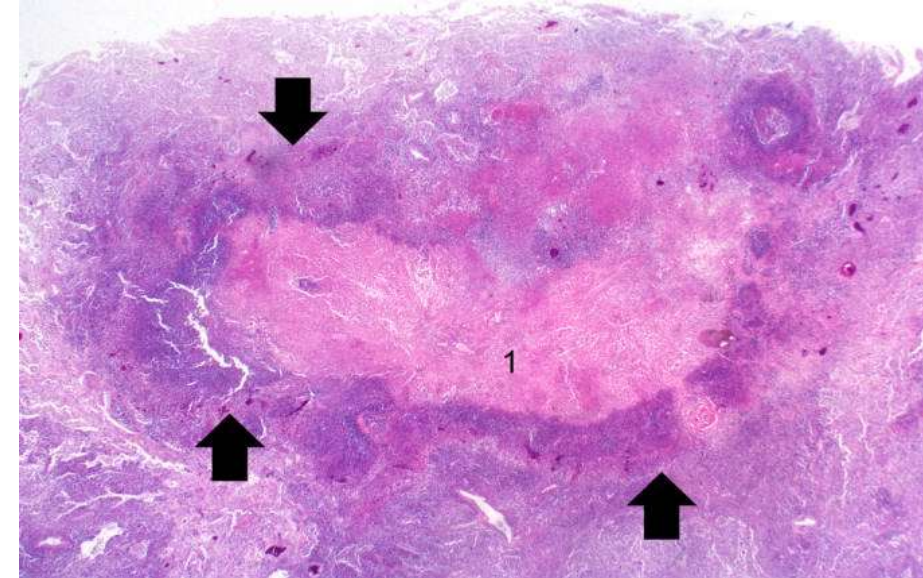
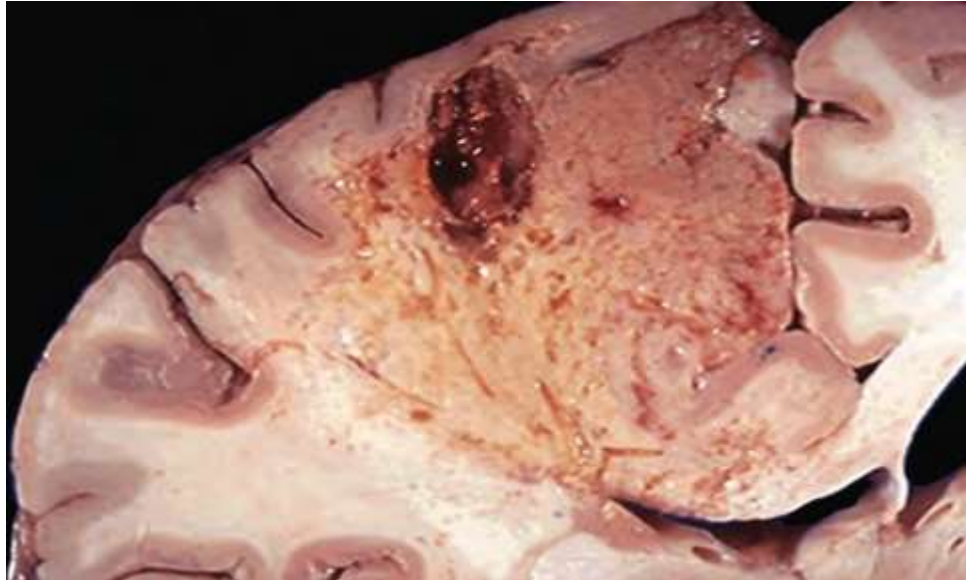
(A) A kidney infarct (yellow)



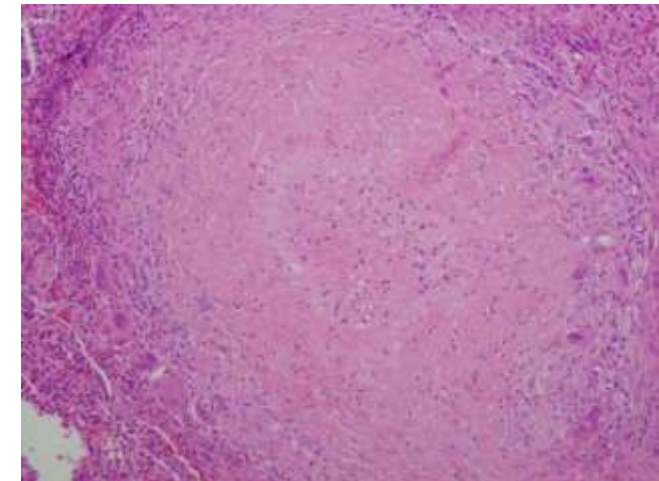
(B) Microscopic view of the edge of the infarct, with normal kidney (N) and necrotic cells in the infarct (I). The necrotic cells show preserved outlines with loss of nuclei,

Cellular Death

- ❑ **Liquefactive necrosis:** Seen in the brain after stroke or in abscesses. The tissue is digested by hydrolytic enzymes, forming a viscous liquid mass.

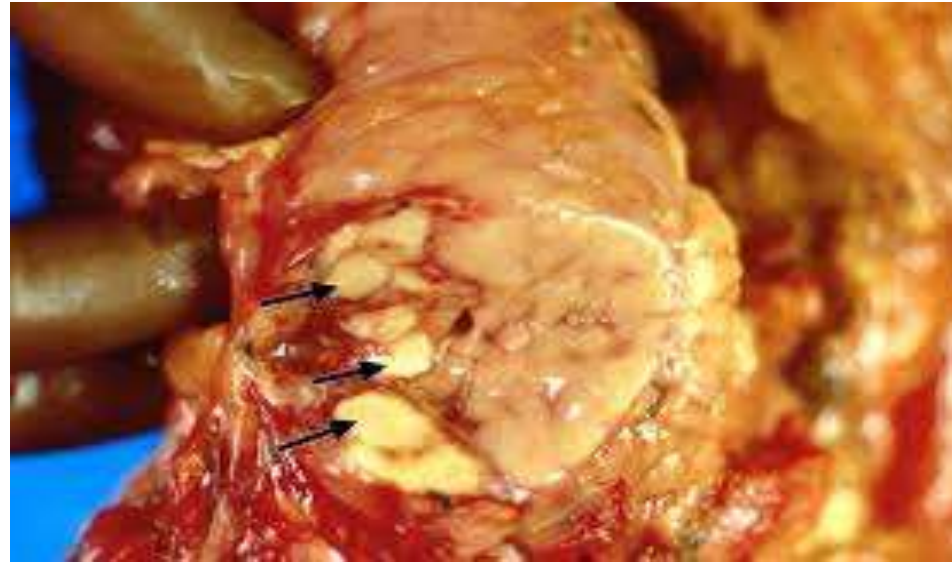


- ❑ **Caseous necrosis:** A cheesy, friable material typically associated with tuberculosis.



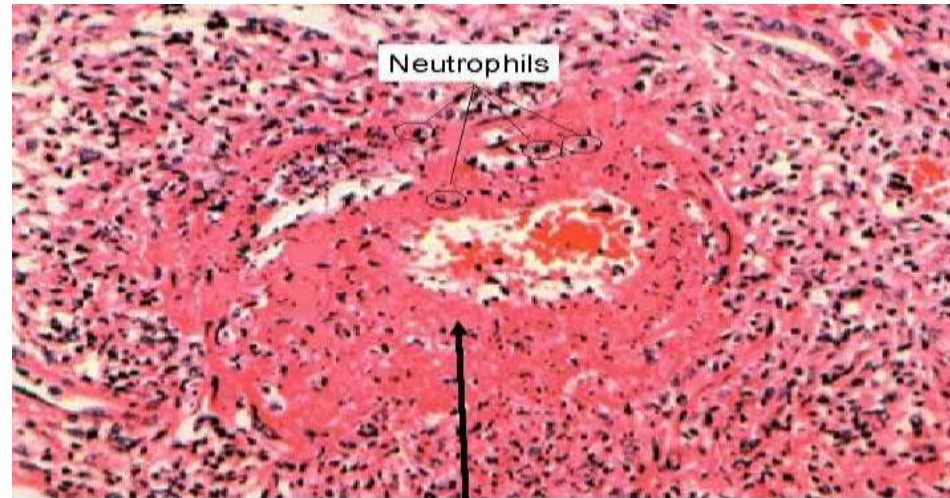
Cellular Death

❑ **Fat necrosis:** Specific to fatty tissue, resulting from the action of lipases (e.g., in pancreatitis)



Fat necrosis in acute pancreatitis.
The areas of white chalky deposits represent foci of fat necrosis with calcium soap formation.

❑ **Fibrinoid necrosis:** is a special form of necrosis. It usually occurs in immune reactions in which complexes of antigens and antibodies are deposited in the walls of blood vessels, but it also may occur in severe hypertension.



Fibrinoid necrosis is a histological finding, as it lacks clear macroscopic signs.

Under a microscope (H&E stain), it's identified by a light pink necrotic area (fibrinoid) around blood vessels, formed by deposited immune complexes and fibrin. Inflammatory white blood cells are visible surrounding this area.

fibrinoid necrosis

Cellular Death

❑ **Gangrenous necrosis** is a term used clinically to describe the necrosis of tissue, typically in a limb (like the fingers, toes, or feet), that has been compromised by ischemia and is often complicated by a superimposed bacterial infection.

❑ 1-Dry Gangrene

Ischemia (Lack of arterial blood supply)

The tissue is **dry, shrunken, and black** (mummified). It results from **coagulative necrosis** where the dead tissue is too low in oxygen and blood to support bacterial growth.

A clear **line of demarcation** forms between the dead and healthy tissue.



Cellular Death

2-Wet Gangrene

Ischemia + Superimposed Bacterial Infection

The tissue is **dark, moist, swollen, soft, and putrid** with a foul odor. It results from a mix of **coagulative** and **liquefactive necrosis** (from the breakdown of tissue by bacteria and inflammatory cells).

No clear line of demarcation is present, and the infection spreads rapidly.



Cellular Death

2B-Gas Gangrene

Infection by gas-forming bacteria (e.g., *Clostridium perfringens*). A severe form of **wet gangrene**.

The tissue is **swollen**, with characteristic **crepitus** (a crackling sound/feeling) on palpation due to gas bubbles produced by the bacteria.



Cellular Death

Universal Microscopic Features of Necrotic Cells

These are the changes seen in the cytoplasm and nucleus of all type of necrotic cells on an H&E stain, regardless of the tissue pattern:

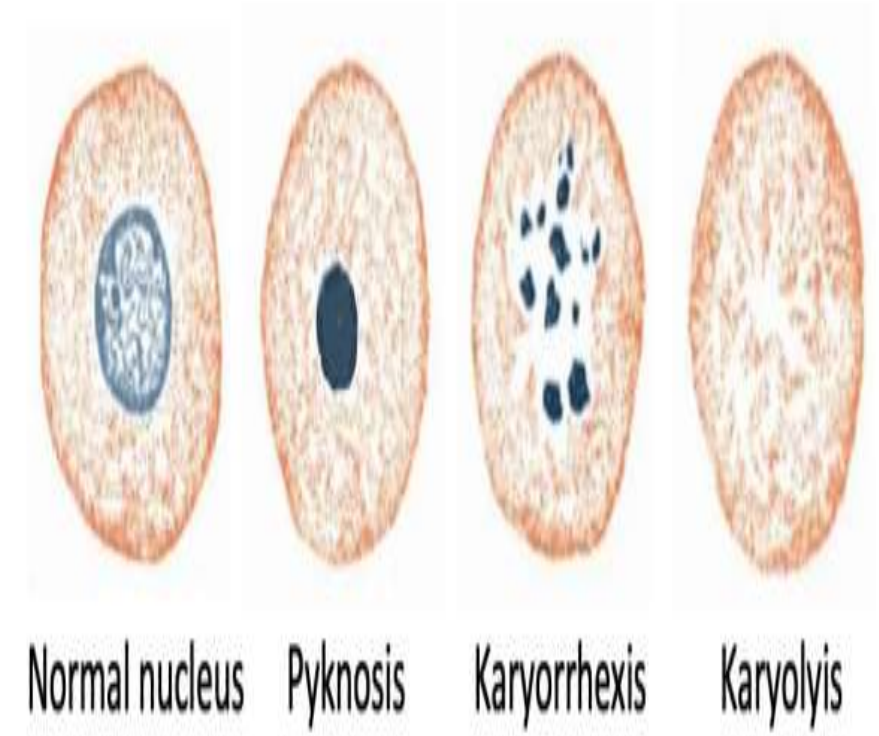
1-Nuclear Changes (The "3 Ks")

These changes reflect the non-specific breakdown of DNA:

A-Pyknosis: The nucleus shrinks and becomes a deeply basophilic (dark blue/black) dense mass of condensed chromatin.

B-Karyorrhexis: The pyknotic nucleus fragments, or breaks up, into small, dark pieces.

C-Karyolysis: The nucleus completely dissolves (fades) as DNase activity digests the chromatin. The nuclear staining disappears, leaving a "ghost" cell.



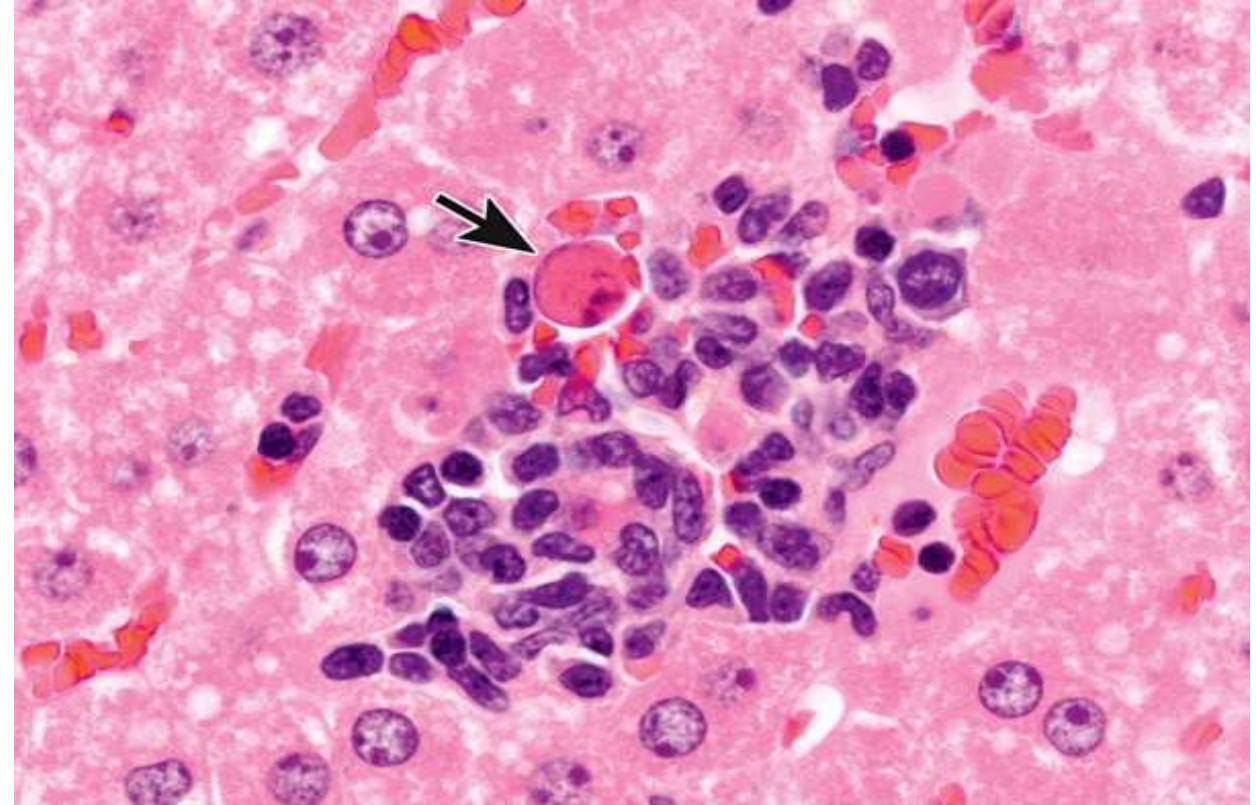
Cellular Death

2-Cytoplasmic Changes

❑ **Increased Eosinophilia:** The cytoplasm appears more intensely pink (eosinophilic) because of:

A-Loss of cytoplasmic RNA (which is normally basophilic/blue),

B-Increased binding of the pink eosin stain to the denatured intracellular proteins.



Cellular Death

B-Apoptosis: Also known as programmed cell death. It's a highly regulated, energy-dependent process that is both physiological (e.g., embryonic development) and pathological (e.g., viral hepatitis).

Apoptosis involves **cell shrinkage**, **membrane blebbing**, and the formation of **membrane-bound vesicles** called **apoptotic bodies**, which are phagocytosed without eliciting inflammation.

Apoptosis

- programmed cell death -

